Two-Layer Salinity Model for SSWD-EvoEpi

WOA23 Climatological Baseline + Fjord Freshwater Depression

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1 Executive Summary

We introduce a two-layer salinity model for the Sea Star Wasting Disease Evolutionary-Epidemiological (SSWD-EvoEpi) simulation framework. The model replaces the previous parametric baseline with real oceanographic data while retaining the ability to represent unresolved fjord-scale freshwater influence.

Layer 1 uses the World Ocean Atlas 2023 (WOA23) monthly surface salinity climatology at 0.25° resolution, providing site-specific seasonal cycles for all 896 simulation nodes. This replaces the latitude-only parametric formula $S_{\text{ocean}}(\phi) = 31.32 + 0.054 \times (\phi - 50)$, which produced zero seasonal variation.

Layer 2 adds a fjord freshwater depression scaled by site enclosedness, latitude-dependent glacial melt, and a seasonal cosine pulse. This captures sub-grid freshwater inputs that WOA23's coarse resolution cannot resolve.

Key improvements over the parametric model:

- **Seasonal realism**: WOA23 mean seasonal range of 1.68 psu (vs. 0.0 psu parametric)
- **Regional accuracy**: overall RMSE reduced from 3.00 to 2.02 psu against DFO lighthouse stations (correlation r improved from 0.519 to 0.660)
- **Timing diversity**: salinity minima occur across all 12 months (not just summer), capturing upwelling-driven minima in winter and rain-driven minima in fall
- **Backward compatible**: when `fw_strength=0`, the model reduces to the WOA23 baseline with no fjord depression

This report presents the model design, WOA23 data coverage, comparison with the parametric baseline, validation against DFO lighthouse observations, and implications for calibration of the SSWD-EvoEpi disease model.

2 Model Design

2.1 Two-Layer Architecture

The salinity at any node i on day-of-year d is:

$$S(i,d) = \max\big(S_{\text{floor}},\ S_{\text{WOA23}}(i,d)\ -\ f_{\text{fw}} \times f_d^{\text{exp}} \times f_{\text{melt}}(\phi_i) \times P(d)\big) \tag{1}$$

where $S_{\text{floor}} = 5.0$ psu is a physical minimum.

2.2 Layer 1: WOA23 Monthly Climatology

The baseline $S_{\text{WOA23}}(i, d)$ is derived from the NOAA National Centers for Environmental Information (NCEI) World Ocean Atlas 2023 monthly surface salinity product. For each of 896 nodes:

1. Find the nearest WOA23 ocean grid cell (0.25° grid).
2. Extract the 12 monthly salinity values.
3. Interpolate linearly between month midpoints to obtain 365 daily values (wrapping December→January for continuity).

Of the 896 sites, 235 fall within a direct WOA23 ocean grid cell; the remaining 661 use nearest-neighbor fill from the land-masked grid. The maximum nearest-neighbor distance is 0.177° (≈19.6 km), ensuring all sites have reasonable ocean salinity estimates.

2.3 Layer 2: Fjord Freshwater Depression

The depression term adds sub-grid freshwater influence for enclosed sites:

$$\Delta S(i, d) = f_{\text{fw}} \times f_d^{\text{exp}} \times f_{\text{melt}}(\phi_i) \times P(d) \tag{2}$$

$$f_{\text{melt}}(\phi) = \text{clamp}\left(\frac{\phi - 35}{25},\ 0,\ 1\right) \tag{3}$$

$$P(d) = \max\left(0,\ \cos\left(\tfrac{2\pi(d-166)}{365}\right)\right) \tag{4}$$

Components:

- f_{fw}: calibration parameter (psu), default 0 (off)
- f_d: `fjord_depth_norm` $\in [0, 1]$ from site enclosedness analysis
- exp: exponent on f_d (default 1.0, linear)
- $f_{\text{melt}}(\phi)$: latitude ramp; zero at 35°N, full at 60°N
- $P(d)$: cosine pulse peaking at day 166 (≈June 15)

2.4 Parametric Fallback

When WOA23 data is unavailable, the model falls back to the original parametric baseline:

$$S_{\text{param}}(\phi) = 31.32 + 0.054 \times (\phi - 50.0) \tag{5}$$

This produces a constant salinity per site with no seasonal cycle.

3 WOA23 Data Coverage

The WOA23 surface salinity product covers the Northeast Pacific from 26.56°N to 61.19°N across 18 subregions (Table 1). The 0.25° grid (≈28 km) provides adequate coverage for open-coast and island sites but necessarily smooths over small inlets and fjords—motivating Layer 2.

Table 1: Regional site counts and WOA23 comparison statistics.

Region	**N sites**	**Bias (psu)**	**RMSE (psu)**	**Lat (°N)**
AK-AL	15	1.12	1.17	52–54
AK-WG	79	−0.17	0.80	51–55
AK-OC	28	−0.05	0.18	57–61
AK-EG	22	−0.27	0.46	58–60
AK-PWS	43	−2.29	2.39	60–61
AK-FN	40	−1.12	1.24	56–59
AK-FS	48	−0.76	1.00	55–57
BC-N	75	−0.17	0.42	50–54
BC-C	36	0.06	0.68	48–51
SS-N	31	−4.75	4.84	48–49
SS-S	89	−3.35	3.44	47–49
JDF	26	−2.37	2.60	48
WA-O	36	−0.04	0.34	46–48
OR	59	1.05	1.30	42–46
CA-N	63	2.20	2.20	36–42
CA-C	97	2.79	2.79	33–36
CA-S	78	3.10	3.10	30–34
BJ	31	3.45	3.46	27–31

The large negative biases in the Salish Sea (SS-N: −4.75 psu, SS-S: −3.35 psu) reflect WOA23 correctly capturing the brackish conditions inside Puget Sound—conditions the parametric model completely missed. Conversely, the positive biases in California/Baja (CA-C: +2.79, BJ: +3.45 psu) show WOA23 capturing the higher salinities of the California Current system.

4 Parametric vs WOA23 Comparison

4.1 Annual Mean Comparison

Figure 1 compares the annual-mean salinity from WOA23 against the parametric baseline across all 896 sites. The overall mean bias is +0.13 psu (WOA23 slightly higher on average) with an RMSE of 2.29 psu and maximum absolute error of 5.74 psu.

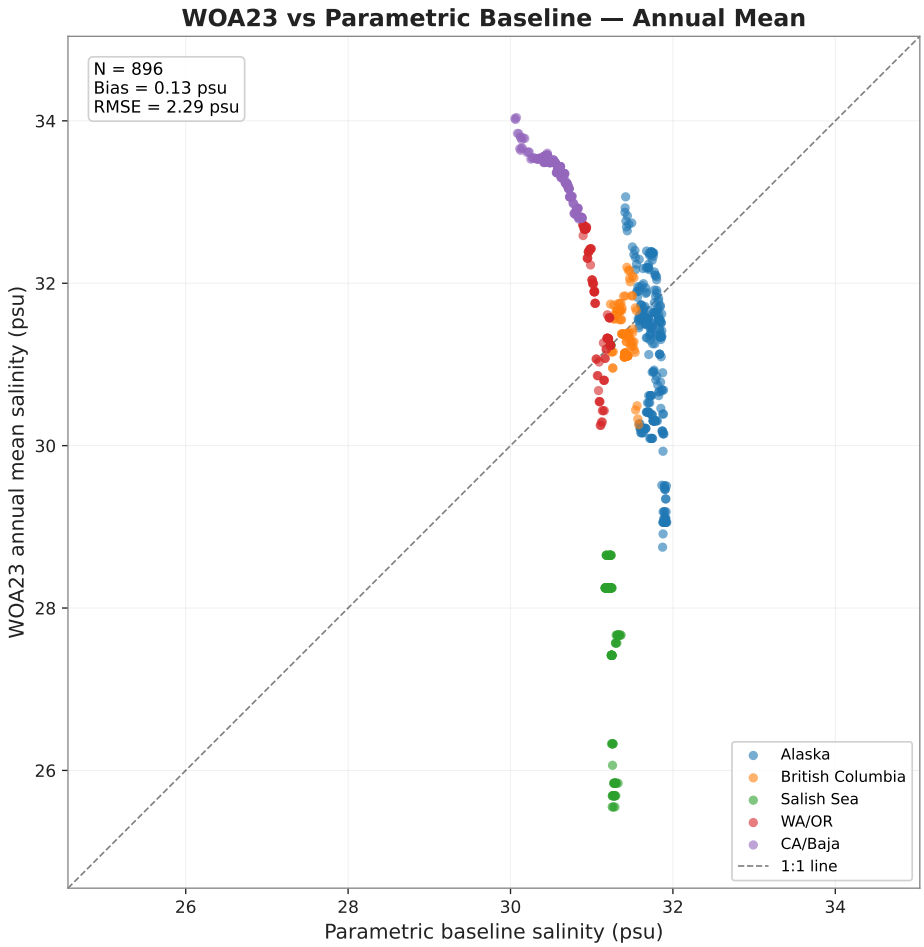

Figure 1: WOA23 annual-mean salinity vs. parametric baseline for 896 sites, colored by macro-region. The 1:1 line (dashed) shows perfect agreement. Scatter reflects both latitude bias and site-specific oceanographic variability the parametric model cannot capture.

The largest systematic deviations occur in two clusters:

- **Salish Sea/JDF** (below 1:1 line): WOA23 is 2–5 psu *lower* than parametric, correctly reflecting the brackish estuarine conditions
- **California/Baja** (above 1:1 line): WOA23 is 2–4 psu *higher*, capturing the elevated salinities of the California Current

4.2 Seasonal Range

The most fundamental improvement is the introduction of seasonal variation. The parametric model produces exactly 0.0 psu seasonal range at every site. WOA23 provides a mean seasonal range (max − min over 12 months) of 1.68 psu, ranging from <0.5 psu at open-ocean sites to >5 psu in fjords and estuaries.

4.3 Timing of Salinity Minima

Figure 2 shows the month of minimum WOA23 salinity across all sites. The salinity minimum is *not* uniformly in summer: while May (171 sites) and June (152 sites) dominate, February (116 sites)

and August (102 sites) are also common, along with significant representation in every month.

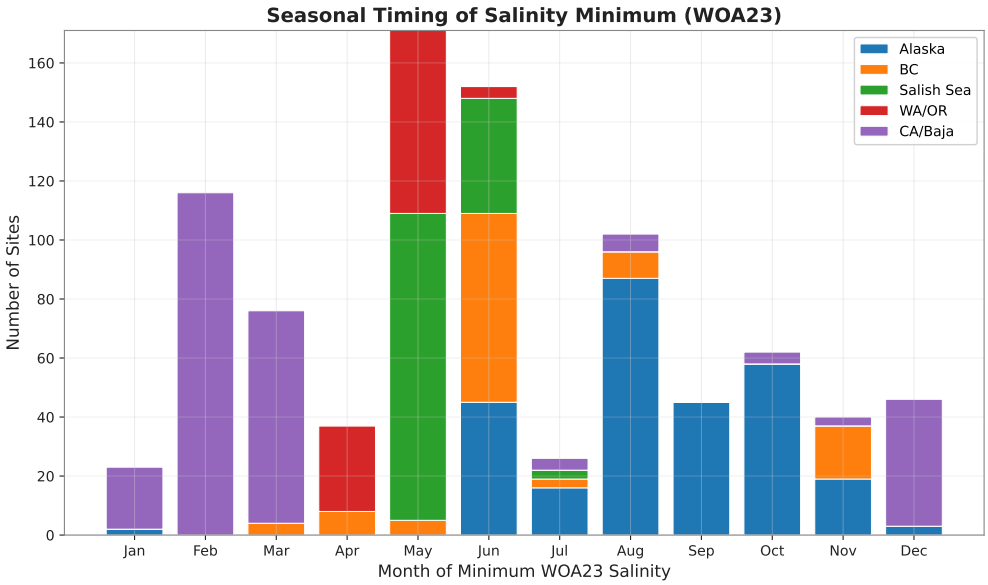

Figure 2: Month of minimum WOA23 salinity across 896 sites, stacked by macro-region. The spread across all 12 months reflects diverse oceanographic drivers: glacial melt (May–Aug in Alaska), winter storms (Jan–Mar in BC), and upwelling relaxation (Aug–Oct in California).

This timing diversity is critical for disease dynamics: the parametric model with its uniform cosine pulse assumed all freshwater influence peaks in June, but WOA23 reveals that salinity minima in California often occur in winter (rain) or late summer (upwelling relaxation)—very different from the glacial melt pattern in Alaska.

5 Combined Model Predictions

5.1 Example Site Profiles

Figure 3 shows the three model layers (WOA23 baseline, parametric baseline, and combined model at `fw_strength=15`) for four contrasting sites spanning the study domain.

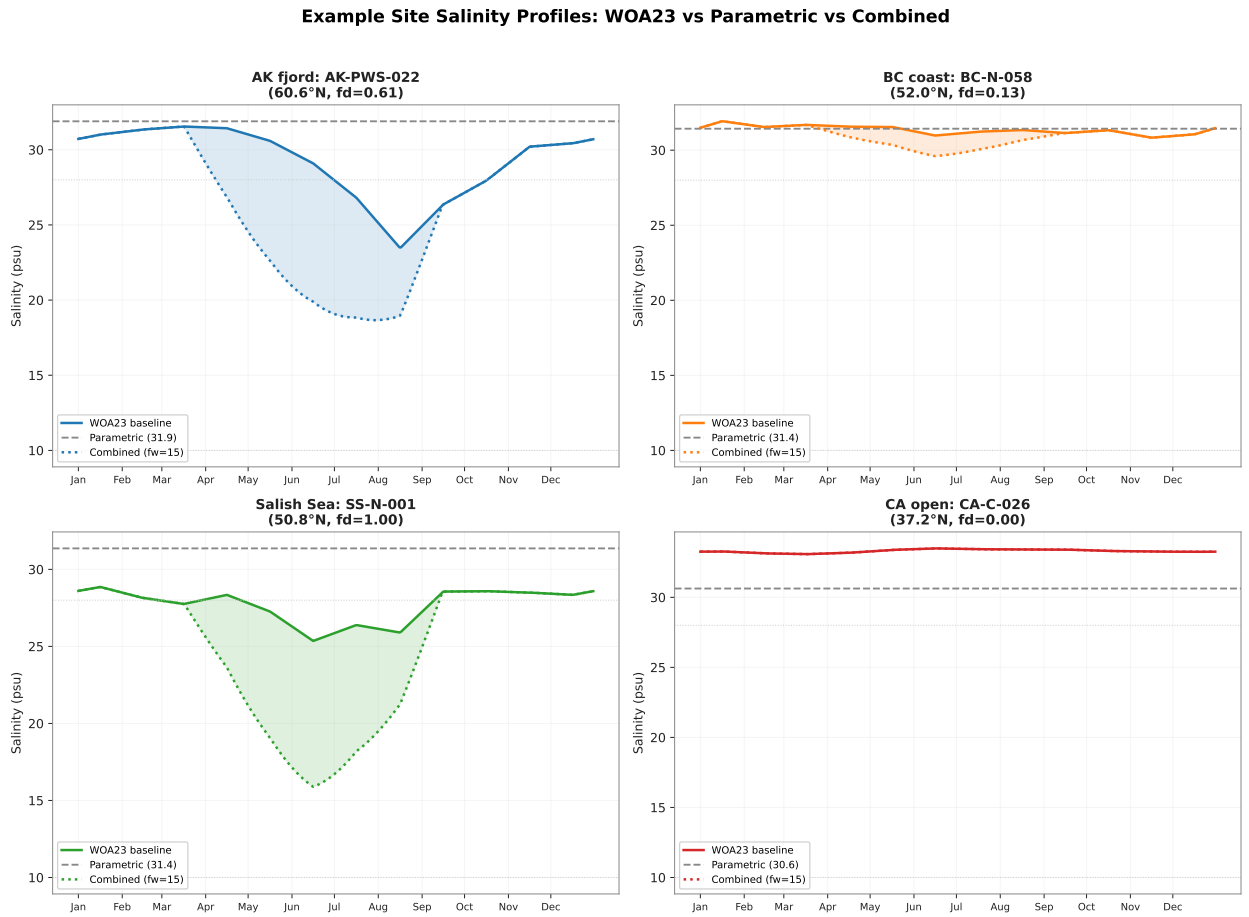

Figure 3: Seasonal salinity profiles at four contrasting sites. Blue solid: WOA23 baseline; gray dashed: parametric (constant); dotted: combined model (`fw_strength=15`). The shaded area shows the fjord freshwater depression from Layer 2. Alaska fjord sites receive maximum depression; California open-coast sites receive none.

5.2 Regional Disease Suppression

Figure 4 presents four-panel regional zoom maps of disease suppression (defined as $1 - \text{sal_mod}$) during June, the peak meltwater month. Suppression depends on both the WOA23 baseline salinity and the Layer 2 fjord depression.

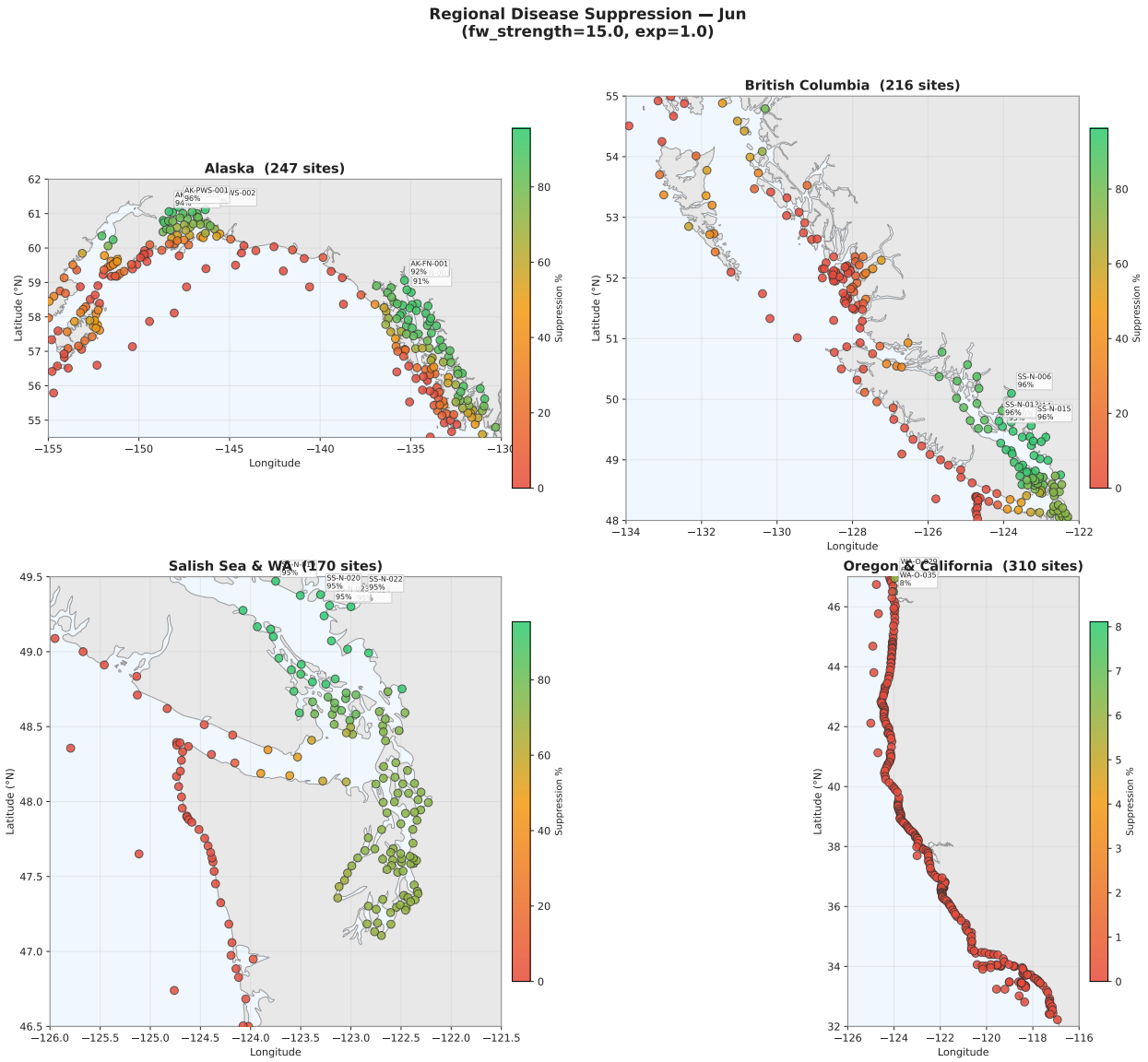

Figure 4: Regional disease suppression from salinity in June (`fw_strength=15`). Green = strong suppression (low salinity), red = no suppression (full-ocean salinity). Coastline polygons from Natural Earth 10m. Alaska shows concentrated suppression in PWS and inner fjords; California shows essentially no suppression.

5.3 Full Coastline Overview

Figure 5 shows the complete study domain from Baja California (26.6°N) to the Aleutian Islands (61.2°N), revealing the strong latitudinal gradient in disease suppression.

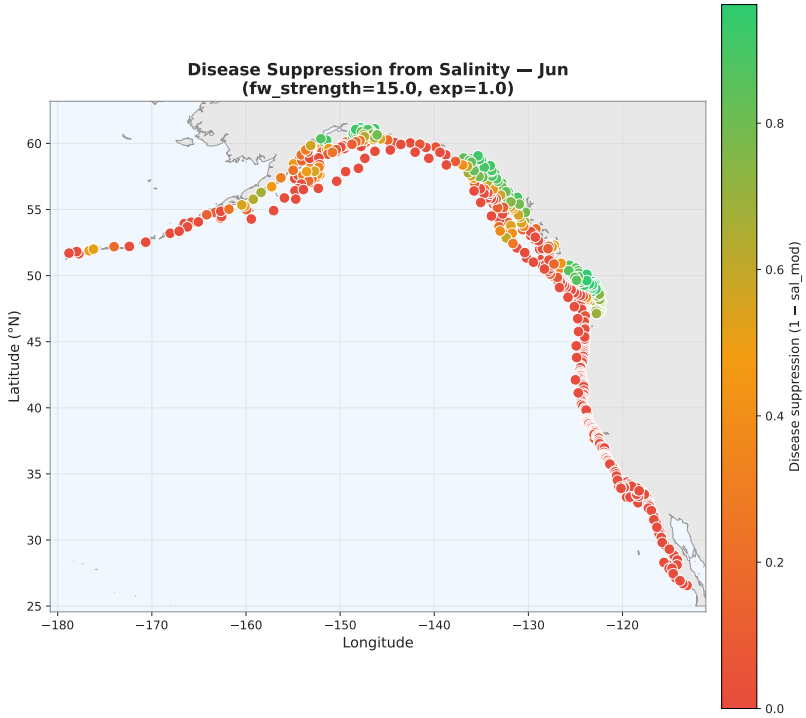

Figure 5: Full-coast disease suppression map, June, `fw_strength=15`. The north-to-south gradient reflects both $f_{\text{melt}}(\phi)$ and the concentration of high-enclosedness sites in Alaska. Sites south of ∼40°N show effectively zero suppression.

5.4 Regional Salinity Profiles

Figure 6 shows the mean seasonal salinity cycle for key regions, illustrating how WOA23 differentiates regions that the parametric model treated identically.

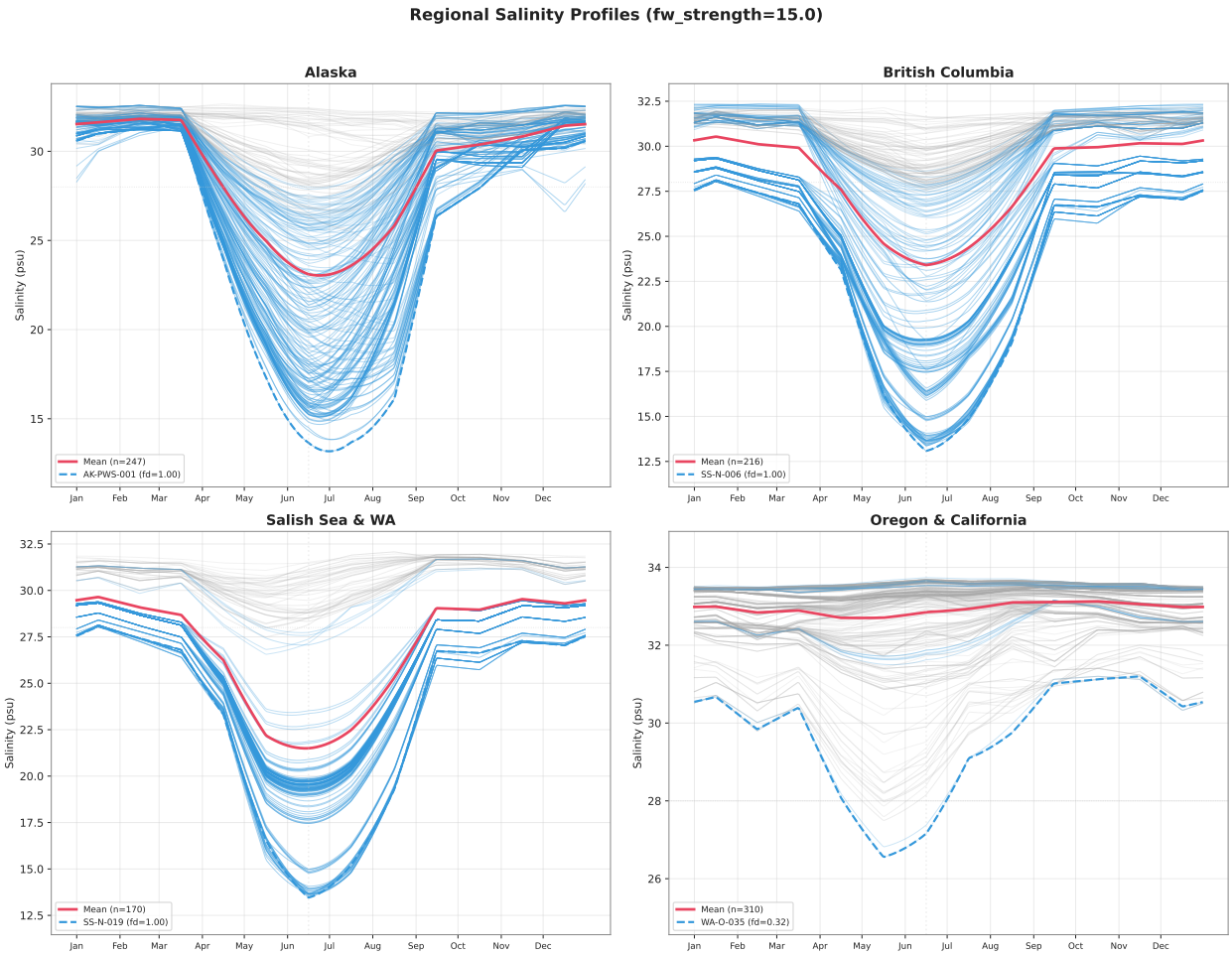

Figure 6: Regional seasonal salinity profiles (WOA23 + fjord depression, `fw_strength=15`). Shaded bands show ±1 SD across sites within each region. Alaska PWS and SE regions show the deepest summer dips; California and Oregon remain near full-ocean salinity year-round.

5.5 Freshwater Strength Sensitivity

Figure 7 shows how varying `fw_strength` from 0 to 30 psu affects salinity and disease suppression at representative sites and across regions.

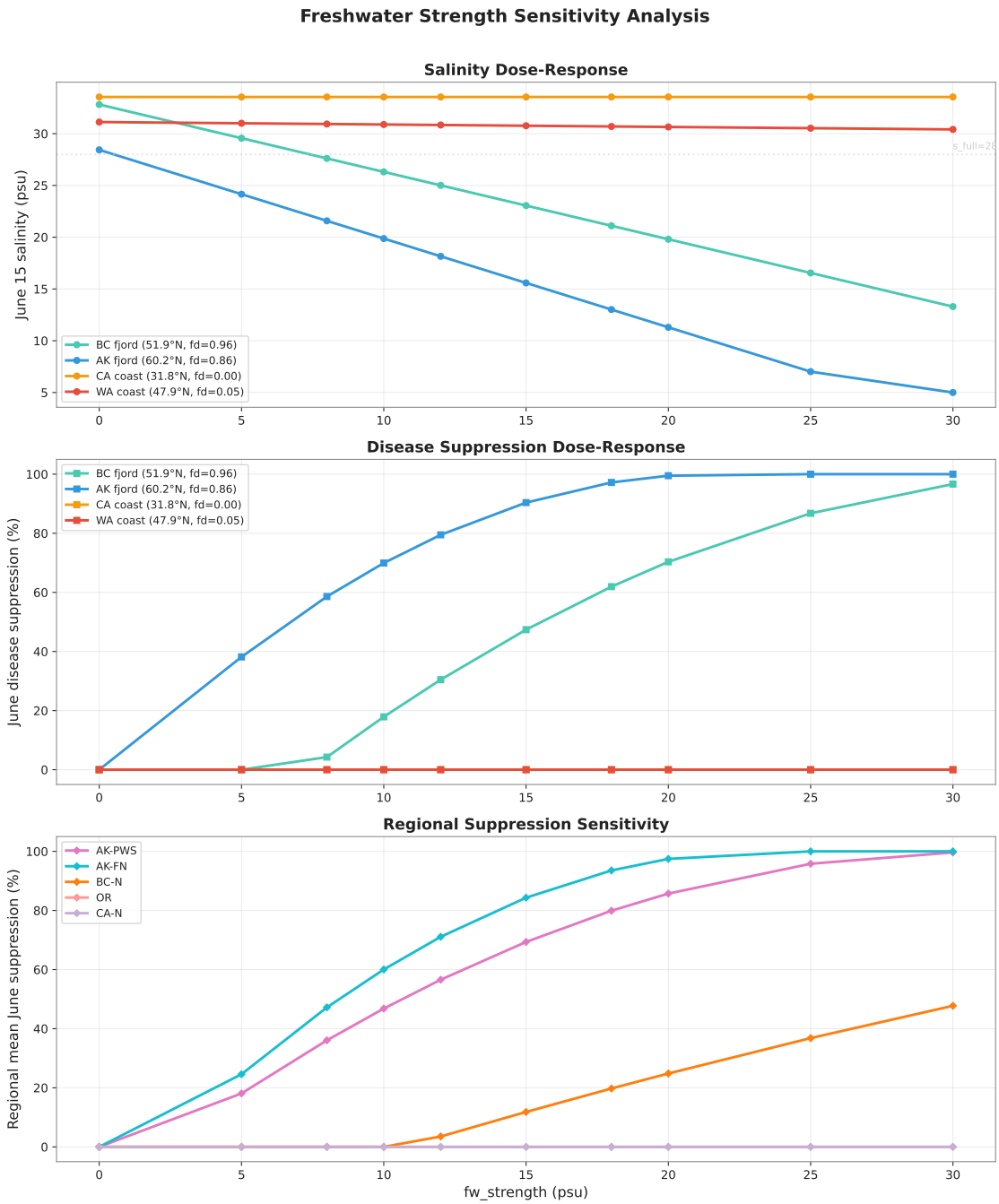

Figure 7: Sensitivity of June salinity and disease suppression to `fw_strength`. Top: June 15 salinity at four representative sites. Middle: corresponding disease suppression (%). Bottom: regional mean suppression. Alaska fjord sites transition to full suppression around `fw=10-15`; California and WA coast sites are unaffected regardless of `fw_strength` (open coast × low latitude = 0 depression).

6 DFO Lighthouse Validation

We validated both models against monthly climatological salinity from 12 Department of Fisheries and Oceans (DFO) lighthouse stations along the British Columbia coast (48.3–54.3°N), providing

144 observation-prediction pairs (12 stations × 12 months).

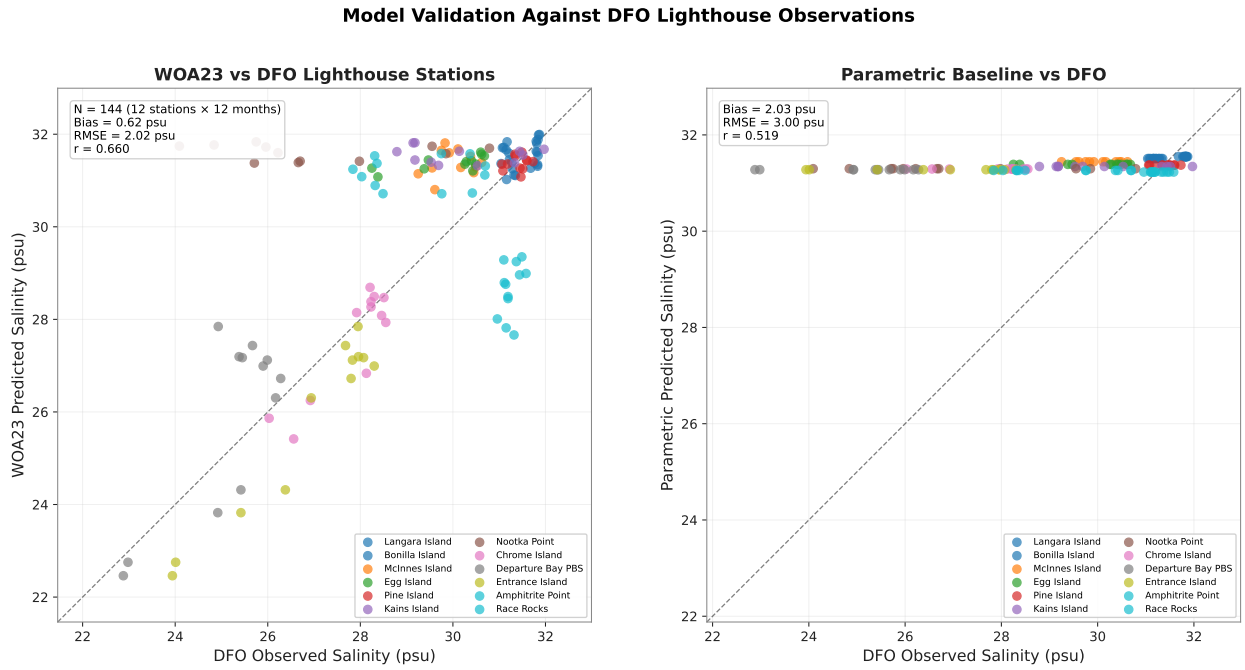

Figure 8: Model validation against DFO lighthouse stations. Left: WOA23 predictions vs. observations. Right: parametric predictions vs. observations. WOA23 shows improved correlation ($r = 0.660$ vs. 0.519) and reduced RMSE (2.02 vs. 3.00 psu). Note that DFO stations are on headlands (`fjord_depth_norm`≈0), so only the WOA23 baseline layer is evaluated here.

Table 2: DFO lighthouse validation summary (12 stations, 144 points).

Metric	**WOA23**	**Parametric**	**Improvement**
Mean bias (psu)	0.62	2.03	69% lower
RMSE (psu)	2.02	3.00	33% lower
Correlation (r)	0.660	0.519	+0.141
Max error (psu)	7.65	—	—

The WOA23 model outperforms the parametric baseline on all three metrics. The remaining RMSE of 2.02 psu is dominated by a few stations with strong local freshwater influence (e.g., Nootka Point, Departure Bay) where even WOA23's 0.25° resolution cannot fully resolve the coastal gradient. These are precisely the sites where Layer 2 (fjord depression) would apply in the full model.

7 Implications for Calibration

The transition from parametric to WOA23 baseline has several important implications for model calibration:

1. **Reduced `fw_strength` requirement**: Because WOA23 already captures much of the coastal freshwater signal (seasonal range 1.68 psu), the Layer 2 depression need only represent the

residual sub-grid freshwater—likely requiring a lower `fw_strength` than the parametric-only model.

2. **Regional timing realism**: The parametric model applied the same June-peaked cosine pulse everywhere. WOA23 provides region-specific seasonal timing automatically. The remaining Layer 2 pulse still peaks uniformly at June 15, but its contribution is smaller, reducing the impact of this assumption.

3. **Salish Sea correction**: The parametric model assigned ∼31 psu to Puget Sound sites (based on latitude alone). WOA23 correctly identifies salinities of 25–28 psu, fundamentally changing the disease dynamics in this critical region.

4. **California salinity increase**: WOA23 places California sites at 33–34 psu (vs. ∼30 psu parametric), reducing any spurious freshwater suppression in the south.

5. **Calibration target**: The `fw_strength` sweep should now explore a range of 0–20 psu (vs. 0–30 psu previously), with the expectation that optimal values will be lower since the baseline already contains seasonal variation.

8 Limitations

1. **Nearest-neighbor fill**: 661 of 896 sites (73.8%) use nearest-neighbor fill from WOA23 ocean cells rather than direct grid-cell extraction. While the maximum NN distance is only 0.177° (≈19.6 km), this means inland and deep-fjord sites receive salinities from the nearest open-ocean cell, potentially overestimating their baseline.

2. **Uniform pulse timing**: The Layer 2 cosine depression pulse peaks at day 166 (June 15) for all sites, regardless of local hydrology. In reality, peak freshwater discharge varies by weeks across the study domain. This is partially mitigated by WOA23 already capturing the site-specific timing in Layer 1.

3. **Resolution limits**: WOA23 at 0.25° (≈28 km) cannot resolve narrow channels, small inlets, or micro-fjords. Sites deep within complex coastlines may have WOA23 values biased toward open-ocean conditions.

4. **Climatological vs. interannual**: WOA23 provides a long-term climatological mean, not year-specific conditions. Anomalous years (e.g., marine heatwaves, strong El Niño) are not represented.

5. **DFO validation geography**: All 12 DFO lighthouse stations are in British Columbia (48–54°N). We lack independent validation data for Alaska and California, the regions where WOA23 diverges most from the parametric model.

Appendix: Model Parameters

Table 3: Model parameters and defaults.

Parameter	**Symbol**	**Default**	**Description**
`fw_strength`	f_{fw}	0.0	Freshwater depression strength (psu)
`fw_depth_exp`	exp	1.0	Exponent on fjord depth norm
`s_floor`	S_{floor}	5.0	Physical minimum salinity (psu)
`s_min`	—	10.0	Vibrio zero-activity threshold (psu)
`s_full`	—	28.0	Vibrio full-activity threshold (psu)
Peak day	—	166	Day of year for melt pulse peak